

Multiple Choice

1. **Answer:** B

Options: A) cell-surface receptors; B) adhesion plaques; C) endosome formation; D) receptor-mediated endocytosis

Note: Adhesion plaques are structures involved in cell adhesion and signaling, but they do not play a direct role in the internalization process of low-density lipoprotein (LDL) cholesterol complexes, making them the correct answer for this question.

2. **Answer:** B

Options: A) histones H₂A, H₂B, H₃, and H₄; B) linker DNA; C) histone H₂A and H₂B only; D) the nucleosome linker; E) histone H₃ only

Note: The correct answer is linker DNA because DNase specifically cleaves the DNA connecting nucleosome dimers, which is the linker DNA, allowing the separation into nucleosome monomers.

3. **Answer:** B

Options: A) Frame shift mutations and nucleotide repeats; B) Base substitutions and small insertions and deletions; C) Base additions and large insertions; D) Silent mutations and missense mutations; E) Duplication mutations and triplet repeat expansions

Note: The correct answer identifies the primary types of point mutations-base substitutions (which alter a single nucleotide) and small insertions and deletions (which change the overall sequence length)-that can significantly affect gene function and protein synthesis.

4. **Answer:** A

Options: A) Temperature, humidity, and air currents; B) Plant species, root depth, and wind speed; C) Soil fertility, irrigation methods, and cloud cover; D) Water salinity, leaf color, and time of day; E) Leaf surface area, stem length, and moon phase

Note: Temperature, humidity, and air currents directly influence the water vapor concentration gradient between the leaf interior and the surrounding environment, thereby affecting the rate at which water evaporates from plant surfaces during transpiration.

5. **Answer:** D

Options: A) Acquired characteristics are inherited through a blending of parental traits; B) Acquired characteristics are inherited but do not influence evolution; C) Acquired characteristics are inherited randomly, regardless of their use; D) Acquired characteristics cannot be inherited; E) Acquired characteristics are inherited based on use and disuse

Note: Lamarck's theory posits that traits acquired during an organism's lifetime can be passed on to its offspring, while current biological evidence demonstrates that genetic inheritance is determined by DNA, which does not include acquired characteristics.

True / False

6. **Answer:** False

Options: A) True; B) False

Note: The quantity of DNA in a eukaryotic cell typically doubles during the S phase of interphase, not during telophase, which is the stage of mitosis where the cell prepares to divide and the DNA is already duplicated.

7. **Answer:** True

Options: A) True; B) False

Note: Proprioceptors are specialized sensory receptors located in muscles, tendons, and joints that detect changes in body position and movement, allowing the brain to coordinate actions without relying on visual information.

8. **Answer:** True

Options: A) True; B) False

Note: This statement is true because all living organisms are composed of cells, which serve as the basic building blocks that perform essential life processes and functions.

9. **Answer:** False

Options: A) True; B) False

Note: The glomerular filtration rate is primarily influenced by factors such as blood pressure and the filtration surface area, rather than the concentration of sodium in the distal tubule and renal blood flow.

10. **Answer:** False

Options: A) True; B) False

Note: The statement is false because autotrophs, which produce their own food through processes like photosynthesis, evolved independently from heterotrophs, which rely on consuming organic materials, and their emergence was influenced by the development of photosynthetic organisms rather than an overabundance of organic nutrients.

Long Form Response

11. **Answer:** The transmission of impulses in plants is evidenced by the physiological responses of the sensitive plant, *Mimosa pudica*, to external stimuli. When the leaves of *Mimosa pudica* are touched, they exhibit rapid folding and drooping, a response that occurs within seconds. This movement is facilitated by changes in turgor pressure within the plant cells, which are triggered by electrical signals that propagate through the plant's vascular system. This phenomenon demonstrates that plants are capable of rapid signaling similar to nerve impulses in animals, indicating a form of communication and response to environmental factors. Overall, the quick reactions of *Mimosa pudica* to touch provide compelling evidence for the existence of impulse transmission in plants. *Note:* correct because it highlights the rapid physiological response of *Mimosa pudica* to stimuli, illustrating the role of electrical signals and changes in turgor pressure in the transmission of impulses within plants.

12. **Answer:** Fermentation plays a critical role in cellular respiration under anaerobic conditions, where oxygen is scarce. During this process, glucose is partially broken down to produce ATP, but more importantly, fermentation regenerates NAD^+ from NADH. This regeneration is essential because NAD^+ is required for glycolysis, the initial step of cellular respiration that produces a small yield of ATP. Without the replenishment of NAD^+ , glycolysis would halt, leading to a complete cessation of energy production. Therefore, while fermentation yields less energy than aerobic respiration, it allows cells to continue producing ATP and survive in low-oxygen environments.

Note: correct because it accurately describes how fermentation allows for the regeneration of NAD^+ from NADH, which is crucial for sustaining glycolysis and thereby enabling continued ATP production in the absence of oxygen.